AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 33 · TEACHER GUIDE

Capstone hypothesis and baseline

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make the difference between a plan and a result explicit.

Use complete baseline rows and honest missing-result fields; reserve fresh classroom cases when published examples are already known.

Keep one habit: Model a planning record → check unsupported claims → assess a fresh baseline and criterion → preserve revisions.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Write a testable capstone **hypothesis** and a credible **baseline**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Fictional data only; reserve the independent task and fresh retry until assessment

Validation influences selection, so its score is not independent final **evidence**. Freeze the chosen procedure before opening the final test.

Opening retrieval

Without opening the old lesson, explain one key idea from Documenting a network experiment. Give an example and a mistake that the idea helps you avoid.

Look for: Record a network experiment with enough detail to replay its calculations

Reference example, not the only acceptable wording: Teacher explanation: Predict max(0, a*x); start a = 1 with no bias; rate 0.1; two full-batch updates; MSE is mean squared **prediction** error.

T1 and T2 train, V1 validates; no shuffling or normalisation. Select lower validation MSE, ties to earlier update. Record the actual Python version, alongside the code and Python float arithmetic.

Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

Look for: Explain how a changed **input** mix can change overall accuracy

Reference example, not the only acceptable wording: Teacher explanation: Clear accuracy is 90 percent in both cohorts; dim accuracy is 50 percent in both.

Development has 80 percent clear and 20 percent dim; new use has 20 percent clear and 80 percent dim. The mix changes while observed group rates remain the same.

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate vector length and normalise a nonzero vector

Reference example, not the only acceptable wording: Teacher explanation: Lengths are 5, 10 and 5. Unit vectors are [0.6, 0.8], [0.6, 0.8] and [-0.6, -0.8].

The length difference between Q and A disappears; the opposite direction of C is preserved.

Pictured end goal

Explain capstone **hypothesis** and **baseline** with a traceable toy-network case and a bounded conclusion.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Documenting a network experiment

Dataset shift in neural models

Similarity and normalisation

For the compact alternative schedule, use two 45-minute sessions. This capstone uses constructed scalar ReLU regression so every calculation is inspectable. It is an experiment in **training** and **evidence**, not a real-world deployment.

Preserve written plans, data roles and raw **prediction**s. Optional local Python requires no external packages or AI account. Do not expose independent-task solutions before collecting work.

Outcomes

Write a testable capstone **hypothesis** and a credible **baseline**

Specify data roles and a selection rule before inspecting results

Calculate **baseline** MSE and distinguish a **success criterion** from a result

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach a complete question-to-evidence plan

> Model baseline arithmetic while preserving unmeasured candidate status.

Main question, full data and baseline ledger

A capstone needs a question that **evidence** can answer. “Build a better network” leaves better undefined.

Here the question is: does two-update full-batch **fitting** of one scalar ReLU weight reduce validation mean squared error relative to the initial-weight **baseline** on a constructed positive-**input** task?

Predict max(0, a*x), with no bias. The **baseline** has a = 1. The trained candidate starts at a = 1, uses learning rate 0.1 and exactly two updates. These choices are written before results are inspected.

The **hypothesis** is that the trained candidate has lower validation MSE. The plan does not guarantee that result. This task tests **training** and record keeping, not whether neural networks are preferable to every other model.

A stronger comparison may also include an ordinary fitted line.

Training rows determine updates. Validation rows compare the prespecified candidate with the **baseline**. The final-test **input**s and targets remain sealed until the procedure and reporting plan are frozen.

MSE = sum of squared **prediction** errors divided by the number of evaluated rows. **Baseline prediction**s on V1 and V2 are 3 and 4. Errors are -3 and -4, squared errors 9 and 16, and validation MSE is 25/2 = 12.5.

There are two validation cases.

Role and ID	x	y
Training T1	1	2
Training T2	2	4
Validation V1	3	6
Validation V2	4	8

Role and ID	x	y
Final test	Sealed	Sealed

Validation ID · Input · Target · Baseline prediction a=1 · Error · Squared error

> **Validation ID:** V1
>
> **Input:** 3
>
> **Target:** 6
>
> **Baseline prediction a=1:** 3
>
> **Error:** -3
>
> **Squared error:** 9

> **Validation ID:** V2
>
> **Input:** 4
>
> **Target:** 8
>
> **Baseline prediction a=1:** 4
>
> **Error:** -4
>
> **Squared error:** 16

> **Validation ID:** MSE denominator
>
> **Input:** Two validation rows
>
> **Target:** (9+16)/2 = 12.5
>
> **Baseline prediction a=1:** Candidate not yet run
>
> **Error:** Final not measured
>
> **Squared error:** No improvement yet established

Main criteria and versioning

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
T1 and T2	Fit two updates	Training result only
V1 and V2	Compare with a = 1	Validation comparison
Sealed final rows	Evaluate frozen procedure	Final evidence on those rows

Each arrow preserves a distinct data role. Opening final rows to choose a learning rate changes their role.

The validation criterion is candidate MSE strictly below **baseline** MSE on V1 and V2. Equality does not pass. The final criterion is lower MSE than the same **baseline** on the sealed final rows.

Report both scores and every raw **prediction** whether or not the criterion passes. Record the rule, **training** rows, update count, learning rate, loss, data versions and code version.

A criterion specifies how to judge a future result; it is not **evidence** that the result already happened.

Keep validation and final criteria separate. Do not silently add epochs or change pre**process**ing after seeing a weak result. A changed plan is a new version and needs an honest account of what **evidence** informed it.

Separate supported baseline key

Guided: errors -3 and -3 give MSE 9 on two rows. Candidate **prediction**s are missing, so the **hypothesis** remains untested.

Exit: final rows became selection **evidence**; document the change, freeze the revised plan and reserve new independent final cases.

Classroom reserved-case handling

Treat the published capstone final cases as teaching material, not a secure assessment bank. Before an individual classroom experiment, hold appropriate fresh evaluation cases outside the learner's planning materials. If learners have already seen future-week final cases or labels, record that exposure and use fresh cases for an independent assessment. A sealed role in a plan does not by itself prevent access.

Separate pictured comparison and missing arithmetic inputs

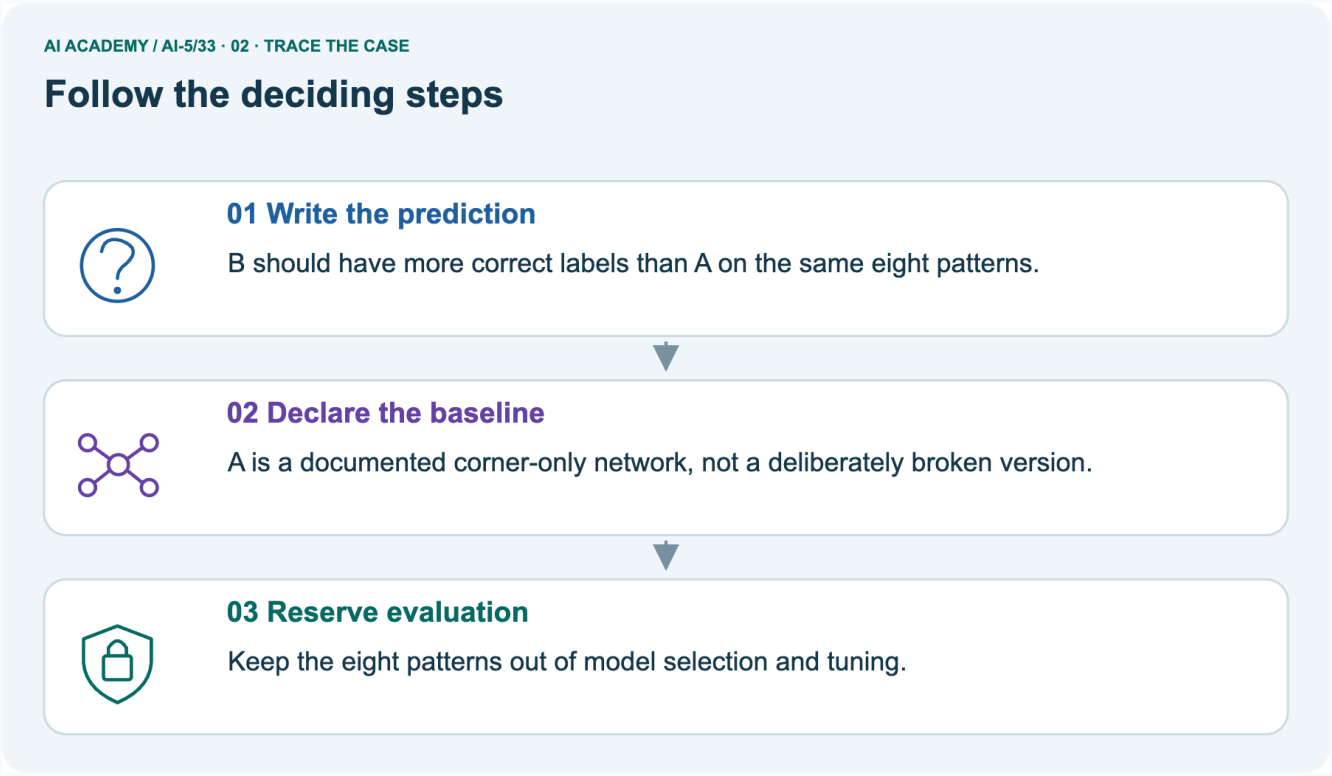
A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

The main capstone is scalar regression with MSE; the picture is a separate proposed image-classification comparison. Pictured A uses corner features; B adds a centre feature. The outcome would be the number correct on the same eight reserved toy patterns.

The pixel grids, model weights, exact labels and predictions are not supplied, so this picture is a planning example, not a runnable scored experiment or evidence that B wins. Both versions must keep reserved labels out of fitting and selection.

Pictured plan and contamination key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Write the prediction	B should have more correct labels than A on the same eight patterns.
2. Declare the baseline	A is a documented corner-only network, not a deliberately broken version.
3. Reserve evaluation	Keep the eight patterns out of model selection and tuning.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A capstone **hypothesis** links a specific architectural change to a measured outcome. A **baseline** anchors the comparison; frozen labels and held-out patterns protect it from answer-key tuning.

Eight toy patterns support a narrow experiment, not claims about real photographs, children or universal architecture superiority.

Change: Give only B the eight evaluation labels during **training**.

Prediction: The proposed comparison becomes contaminated rather than a valid held-out test.

Reason: One version has gained privileged information from the reserved evaluation set.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Hypothesis	A testable expectation with a stated comparison and outcome
Baseline	A specified reference procedure evaluated on the same cases
Success criterion	A rule written before evaluation for judging the measured result
Reserved test	Cases kept out of fitting, tuning and selection

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

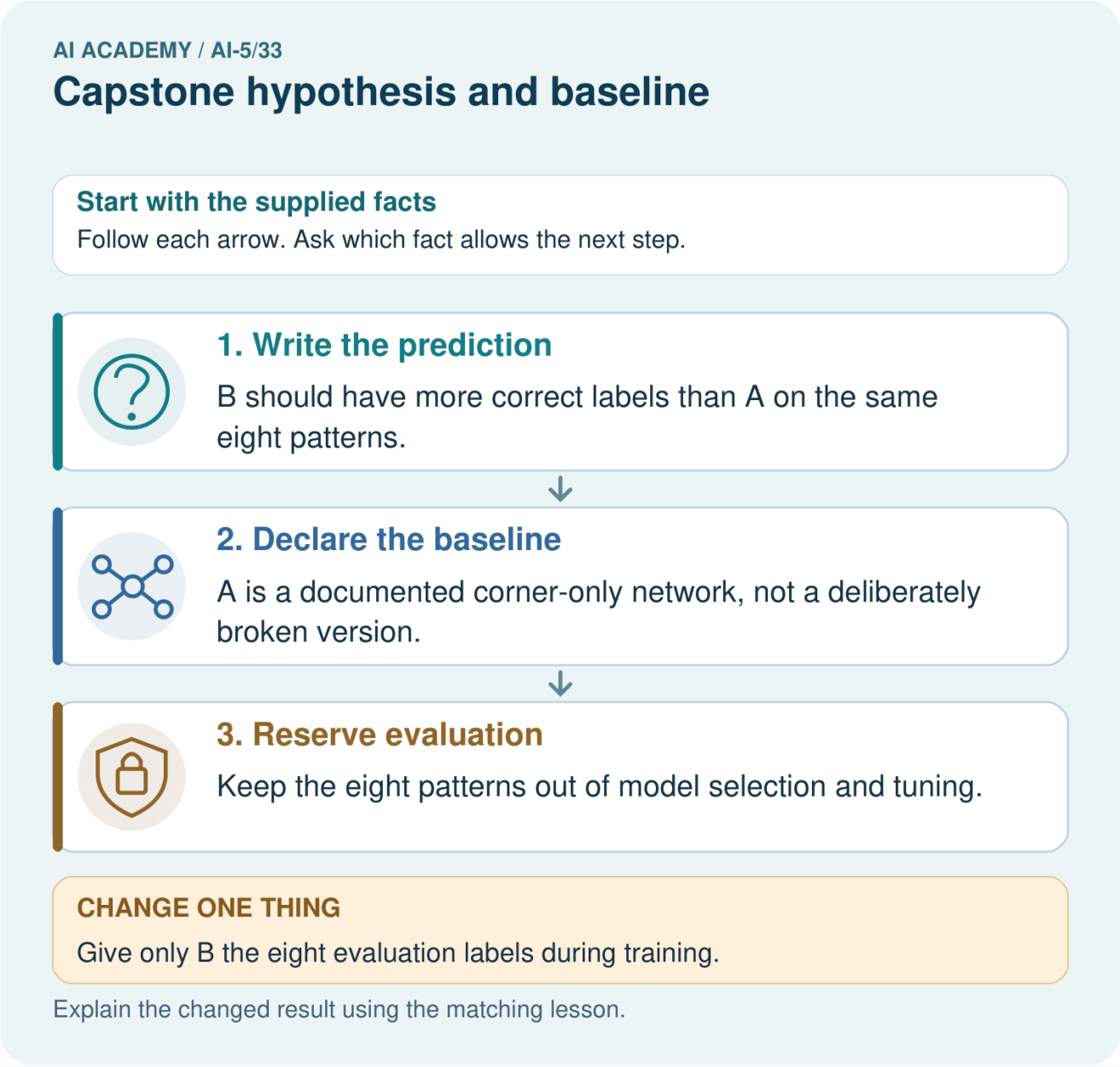
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Capstone hypothesis and baseline

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Compute **baseline prediction**s, errors, squared errors and MSE on V1 and V2. State the denominator.

Teacher check: Teacher explanation: **Prediction**s 3 and 4; errors -3 and -4; squared errors 9 and 16; MSE 12.5 over two rows.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Capstone **hypothesis** and **baseline**. Proposed explanation: “A **hypothesis** is a result”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A **hypothesis** states an expectation before the comparison. Results may support or contradict it.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Name the intended change, the **baseline** and the measured outcome. Why must both versions avoid **training** on the eight reserved labels?

The change is adding the centre feature; A is corner-only. The outcome is correct labels out of the same eight patterns. Using reserved labels to train B leaks evaluation information and invalidates the unseen comparison.

Changed-case reasoning: The proposed comparison becomes contaminated rather than a valid held-out test. One version has gained privileged information from the reserved evaluation set.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a frame with question, fixed settings, changed setting, data roles, raw **prediction**s, metric, conclusion and limitation. Allow a calculator and explicit squared-error columns.

Extension: implement the arithmetic locally, preserve the program version, and explain why a successful replay adds no independent test cases.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and ask students to explain the data roles.
7 to 18 minutes	Read section 1 and complete W1 with a recorded reason.
18 to 32 minutes	Trace section 2 and complete W2; compare raw values before conclusions.
32 to 45 minutes	Explain the diagram and complete W3 in pairs; record any support.
Session 2 start to 8 minutes	Complete the guided response and discuss why the evidence has limits.
8 to 20 minutes	Read the remaining explanation and prepare a concise experiment record.
20 to 35 minutes	Collect W4 independently. Require calculations and a bounded claim.
35 to 45 minutes	Use W5 and the exit response for feedback. Administer the fresh retry after feedback without the earlier solution.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main baseline MSE 12.5; strict candidate criterion lower than 12.5; no result yet. Supported example baseline MSE 9 and missing candidate predictions. Independent baseline a1 on (3, 9), (4, 12) predicts 3/4, errors−6/−8, squares 36/64, MSE 50. Candidate starts 1, rate.1, two full-batch updates; strict validation criterion MSE<50. Success untested until candidate outputs supplied.

Leaked final labels become development evidence; document and reserve new final cases.

Diagnose the current misconceptions

A **hypothesis** is a result: A **hypothesis** states an expectation before the comparison. Results may support or contradict it.

A weak **baseline** proves neural networks help: An initial-weight reference checks improvement from that starting point, not superiority to a fitted line.

Renaming a used test restores independence: A new name cannot undo using its results for selection.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A new task predicts max(0, a*x), with **baseline** a = 1. **Training** rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed.

Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

Teacher explanation: **Baseline prediction**s 3 and 4, errors -6 and -8, squared errors 36 and 64, MSE 50 over two rows.

Candidate is fixed at two updates with rate 0.1 on **training** only; success means candidate validation MSE < 50. No candidate results have been supplied, so success is unknown.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Write the hypothesis

State the **prediction** rule, **baseline**, candidate learning rate and update count, comparison data and metric. Write a testable **hypothesis**.

Teacher explanation: Predict max(0, a*x) without bias; **baseline** a = 1; candidate starts at 1, learning rate 0.1, two full-batch updates on T1/T2.

Hypothesis: candidate validation MSE on V1/V2 is strictly below the **baseline** MSE.

W2 Calculate the reference score

Compute **baseline prediction**s, errors, squared errors and MSE on V1 and V2. State the denominator.

Teacher explanation: **Prediction**s 3 and 4; errors -3 and -4; squared errors 9 and 16; MSE 12.5 over two rows.

W3 Audit the plan

A proposal says “try models until the test score looks good”. Identify the missing fixed choices and explain why that test is not independent.

Teacher explanation: It omits architecture, **baseline**, **training** settings, metric, stopping/selection rule and split policy.

Repeated choices based on the test use it for development; reserve new independent final data after freezing the revised procedure.

W5 Preserve the reserved evidence

A partner inspects final labels and changes the learning rate. Explain how to document this and obtain independent **evidence**.

Teacher explanation: Record the inspection and changed plan. Those rows informed selection and cannot honestly serve as untouched final **evidence**.

Freeze the revised procedure and use newly reserved independent cases; retain the earlier result as development **evidence**.

▼ **Earlier wording, alternatives and picture transcripts**

Guided and exit questions

Guided: errors -3 and -3 give MSE 9 on two rows. Candidate **prediction**s are missing, so the **hypothesis** remains untested.

Exit: final rows became selection **evidence**; document the change, freeze the revised plan and reserve new independent final cases.

W1 Write the hypothesis

W2 Calculate the reference score

W3 Audit the plan

W4 Plan a new case independently

A new task predicts max(0, a*x), with **baseline** a = 1. **Training** rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed.

Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

Teacher explanation: **Baseline prediction**s 3 and 4, errors -6 and -8, squared errors 36 and 64, MSE 50 over two rows.

Candidate is fixed at two updates with rate 0.1 on **training** only; success means candidate validation MSE < 50. No candidate results have been supplied, so success is unknown.

W5 Preserve the reserved evidence

Independent assessment and fresh retry

W4 requires **baseline** MSE 50, all fixed settings, a strict < 50 criterion and no premature success claim. Fresh retry: use validation (3,12) and (4,16), same **baseline** and candidate settings.

Baseline errors -9 and -12 give MSE (81+144)/2 = 112.5; require a < 112.5 criterion and an untested conclusion.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: A new task predicts max(0, a*x), with **baseline** a = 1.

Training rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed. Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates.

Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

Keep actual **observation**s separate from expected results, preserve failures, compare a **baseline** and explain the limits.

Provide a frame with question, fixed settings, changed setting, data roles, raw **prediction**s, metric, conclusion and limitation. Allow a calculator and explicit squared-error columns.

Extension: implement the arithmetic locally, preserve the program version, and explain why a successful replay adds no independent test cases.

Week 34 runs controlled learning-rate comparisons while preserving the recorded data roles.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Delayed return and support record

Suggested return points in this level: ai-5/34, ai-5/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Write a testable capstone **hypothesis** and a credible **baseline**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Compute **baseline prediction**s, errors, squared errors and MSE on V1 and V2. State the denominator.

Reasoned practice answer: Teacher explanation: **Prediction**s 3 and 4; errors -3 and -4; squared errors 9 and 16; MSE 12.5 over two rows.

Check every learner before discussion: Use the worked case for Capstone **hypothesis** and **baseline**. Proposed explanation: “A **hypothesis** is a result”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A **hypothesis** states an expectation before the comparison. Results may support or contradict it.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Give only B the eight evaluation labels during **training**. Require a **prediction** and a reason before comparing. Expected result: The proposed comparison becomes contaminated rather than a valid held-out test.

Reason: One version has gained privileged information from the reserved evaluation set.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Capstone hypothesis and baseline: the case

AI ACADEMY / AI-5/33 · 01 · LOOK AT THE FACTS

Capstone hypothesis and baseline: the case

The situation

A testable image-pattern hypothesis. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

Supplied evidence

Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

Explain capstone hypothesis and baseline with a traceable toy-network case and a bounded conclusion.

Follow the labelled picture with the matching explanation. Open the vector diagram

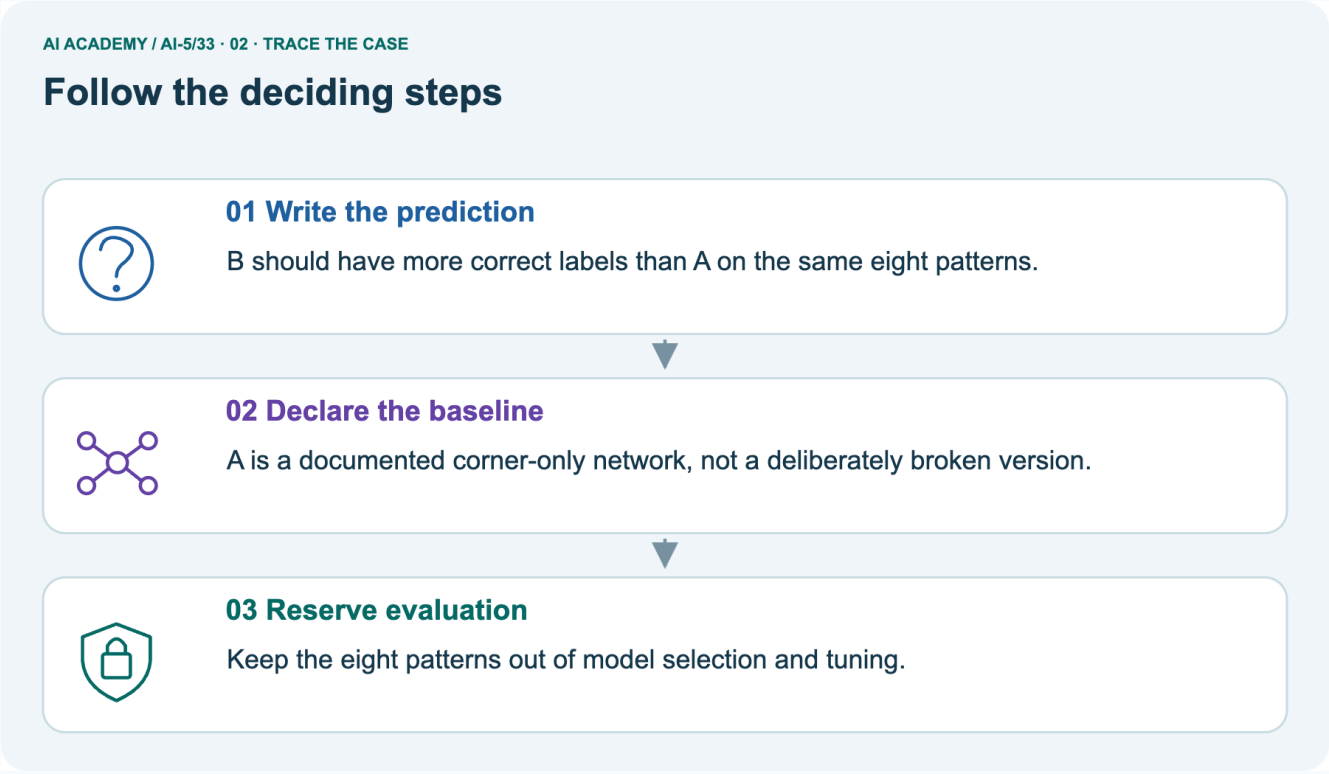
A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

Supplied facts: Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



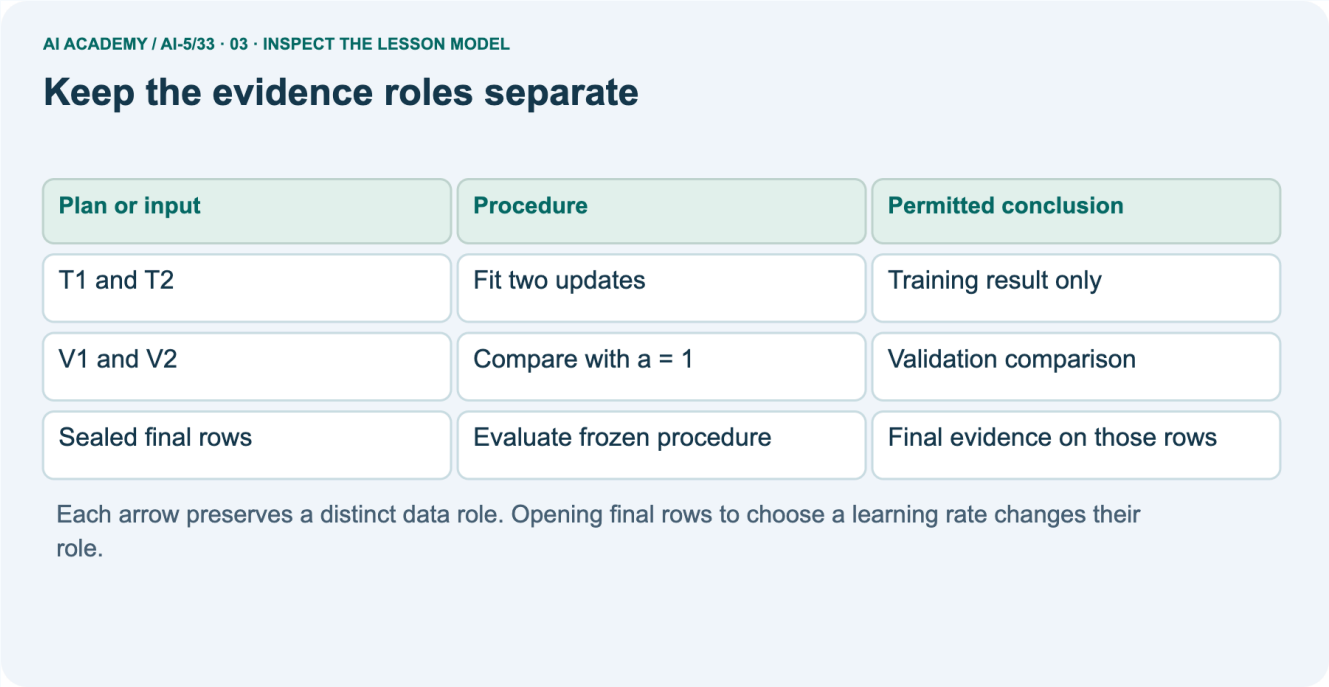
Follow the labelled picture with the matching explanation. Open the vector diagram

Write the **prediction**: B should have more correct labels than A on the same eight patterns. → Declare the **baseline**: A is a documented corner-only network, not a deliberately broken version. → Reserve evaluation: Keep the eight patterns out of model selection and tuning.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Keep the evidence roles separate



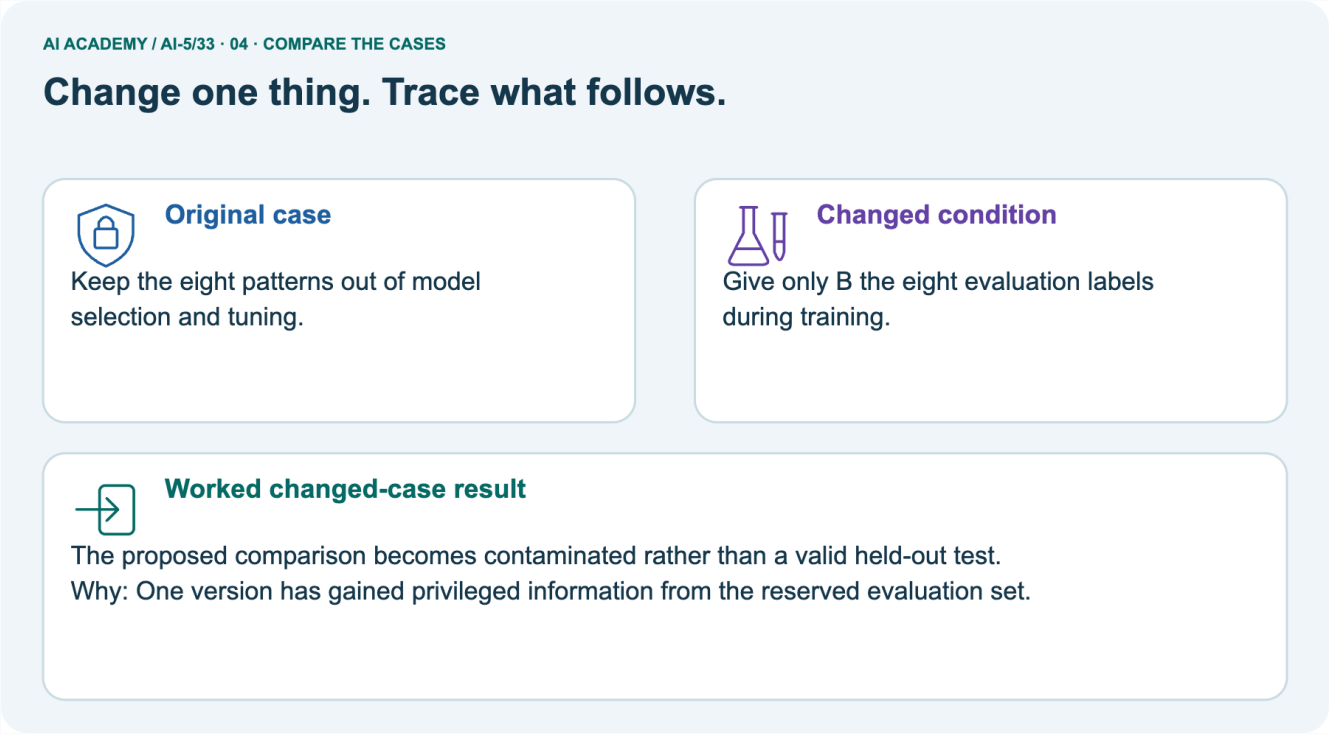
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the **evidence** roles separate. T1 and T2 | Fit two updates | **Training** result only. V1 and V2 | Compare with a = 1 | Validation comparison. Sealed final rows | Evaluate frozen procedure | Final **evidence** on those rows

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Keep the eight patterns out of model selection and tuning. Changed condition: Give only B the eight evaluation labels during **training**.

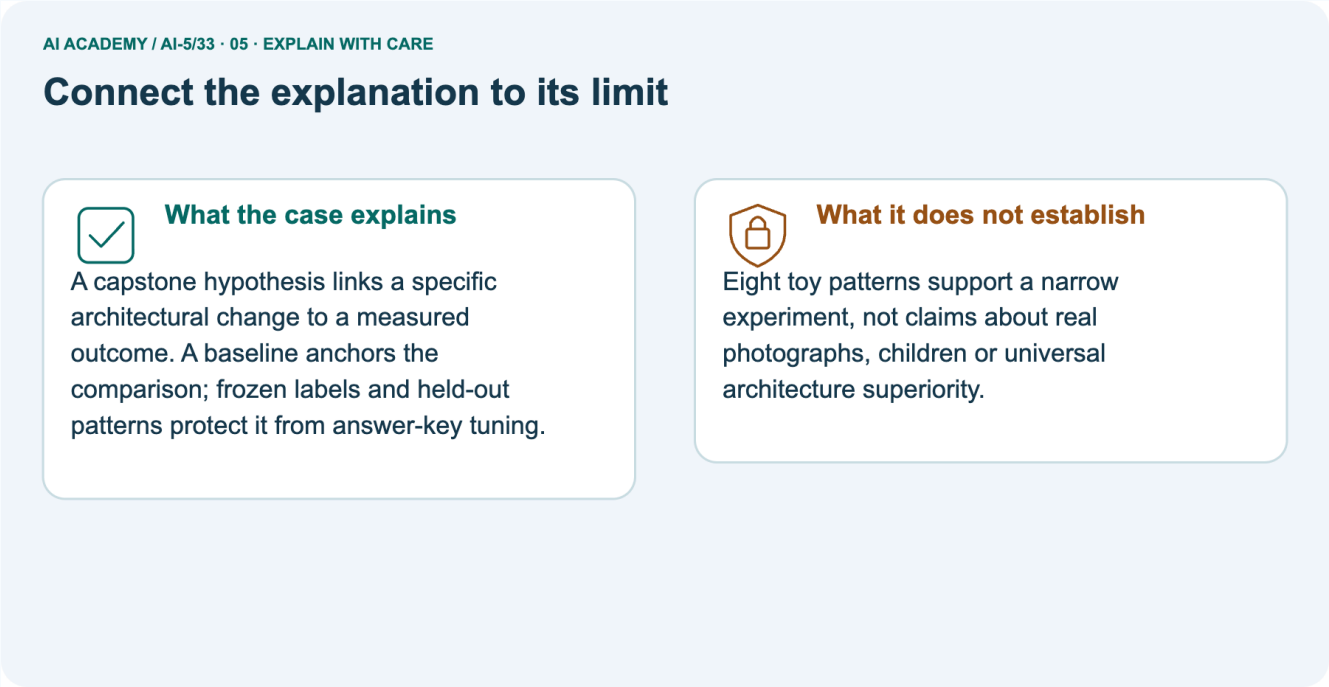
Expected result: The proposed comparison becomes contaminated rather than a valid held-out test. Reason: One version has gained privileged information from the reserved evaluation set.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: One version has gained privileged information from the reserved evaluation set.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A capstone **hypothesis** links a specific architectural change to a measured outcome. A **baseline** anchors the comparison; frozen labels and held-out patterns protect it from answer-key tuning.

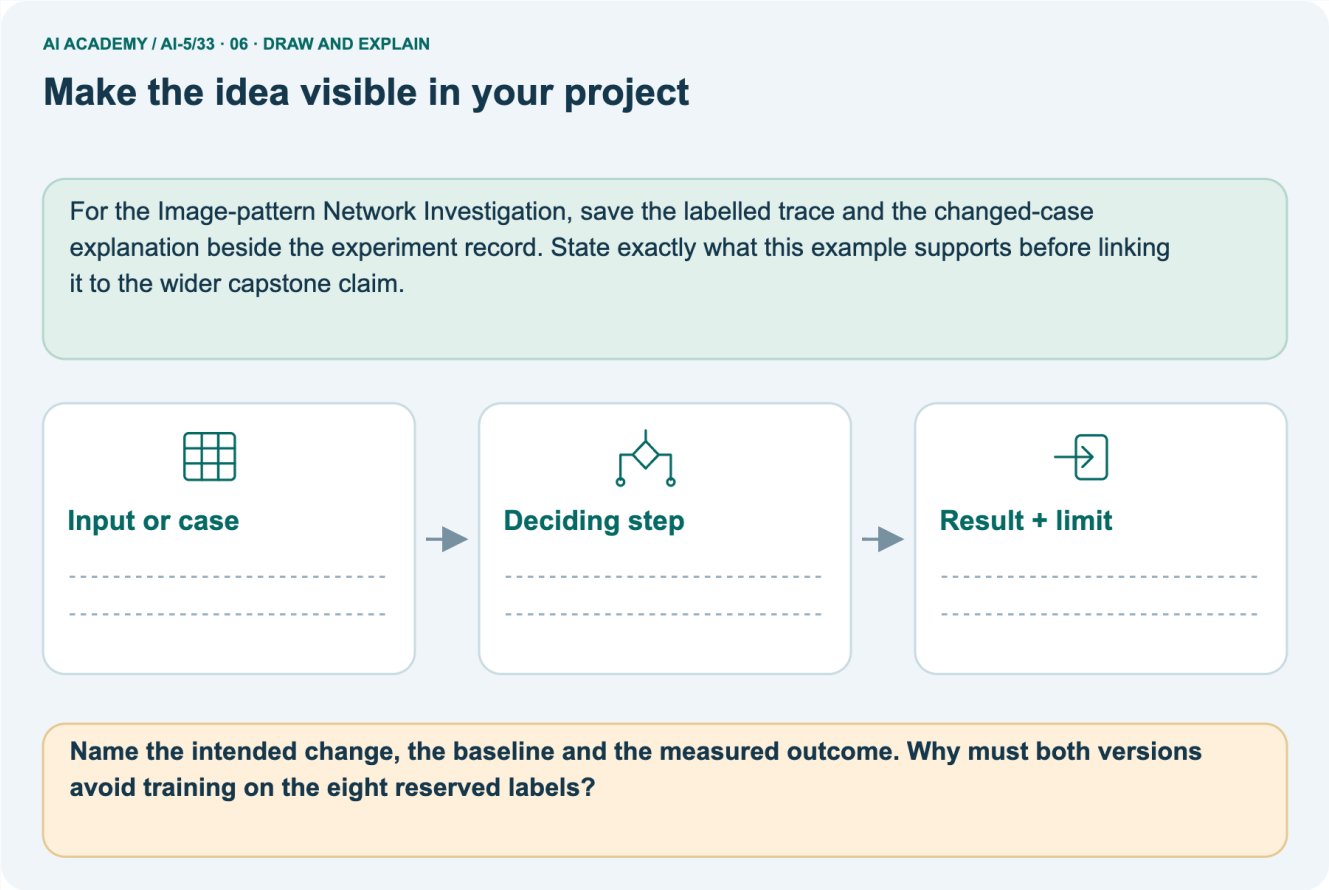
Limit: Eight toy patterns support a narrow experiment, not claims about real photographs, children or universal architecture superiority.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Eight toy patterns support a narrow experiment, not claims about real photographs, children or universal architecture superiority.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Name the intended change, the **baseline** and the measured outcome. Why must both versions avoid **training** on the eight reserved labels?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The change is adding the centre feature; A is corner-only. The outcome is correct labels out of the same eight patterns.

Using reserved labels to train B leaks evaluation information and invalidates the unseen comparison.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Capstone hypothesis and baseline: the case

02. Follow the deciding steps

03. Keep the evidence roles separate

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.